Jesslyn Devina

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Education

BASc in Engineering Physics, *University of British Columbia* 09/2020 – 05/2025

Business Administration, 08/2023 – 12/2023
University of Hong Kong (Exchange Semester)

Technical Experience

Data Engineer, *Insurance Corporation of British Columbia* 05/2024 – 08/2024

- Built SQL and Python ETL pipelines to aggregate 100k+ records, implementing automated data-integrity checks that ensured 100% accuracy for high-stakes risk tradeoff modeling and executive decision-making
- Conducted a 10-year longitudinal study of crash data to isolate high-risk variables; synthesized findings into strategic data visualizations that served as the primary evidence base for provincial safety strategies.
- Developed geospatial traffic simulations to model predictive outcomes for road interventions; engineered custom network configurations and calibrated parameters to reflect localized traffic flow

Product Manager, *Walnut App, New Venture Design UBC* 10/2024 – 05/2025

- Owned the end-to-end UI/UX lifecycle and product design, translating high-fidelity Figma designs into responsive, production-ready code while ensuring technical feasibility
- Conducted 80+ user interviews to identify pain points for an AI MVP, sythesizing qualitative feedback and user stories into testable BDD specifications that guided the build
- Ran the team's sprints and Jira backlog, deconstructing the product roadmap into clear, actionable tasks for the team

Engineering Researcher, *National Research Council Canada* 01/2022 – 04/2022

- Optimized the CCAT-Prime optical design by analyzing lens configurations against mapping speed; demonstrated that a fifth lens caused a 4% throughput loss that outweighed potential gains, preventing unnecessary complexity for the 850 GHz module
- Developed a data pipeline to convert raw sensor signal (IQ timestreams) into frequency shifts and power-absorbed metrics, validating calibration methods for published research
- Redesigned a 100mK coax head in SolidWorks, increasing SMA connector density to expand detector capacity and increase the instrument's mapping resolution

Projects & Involvement

SeedBot , *Autonomous seed scanning system with Insporos*

- Integrated Zaber gantry motion control with a custom optical height-detection system, synchronized via Raspberry Pi to automate high-precision seed positioning
- Designed a non-contact laser triangulation system for a seed-scanning prototype to maintain precise focal distances for 4mm seeds, automating a previously manual inspection process
- Performed root cause analysis on prototype calibration errors, identifying pixel-to-mm scaling discrepancies and line-width noise to provide a technical roadmap for sub-millimeter resolution

Solar Chapter Canada , *Project Coordinator* 01/2024 – 08/2024

- Coordinated cross-functional teams and managed stakeholder alignment for complex infrastructure projects serving 1000+ residents
- Designed a solar-powered pump system providing water access for 1000+ residents

TraffiBot , *ROS-based ML Autonomous Traffic Control Robot*

- Designed and trained a CNN model for license plate recognition, achieving high accuracy via data augmentation and optimization
- Developed an autonomous traffic control robot using ROS and computer vision for real-time traffic control